

Candidate number **B6**

Fraction

The **32nd** Republic of China Biology Olympiad National Team Selection Camp , **2021**

Microbiology and Microbial Ecology Practice

Materials and experimental equipment:

A. Experimental	Quantity B.	Drug solution: 2	quantity
Equipment: 1. Glass slides	tablets.	Incubate at 42°C. Yogurt plus probiotic sample. Item (0hr) (place on ice)	1 tube
2. p20 micropipettes	1 vial 2.	Incubated at 42° C with yogurt and probiotics. Serving (3 hours) (on ice)	1 tube
3. p20 micropipette tip	1 box of	3. Incubated at 42°C with yogurt and probiotics . Serving (6 hours) (placed on ice)	1 tube
4. Lens cleaning paper	1 packet	4. Incubate at 42°C with yogurt and probiotics. Serving (9 hours) (on ice)	1 tube
5. Dust-free paper	1 box of	5. Crystal violet staining solution	1 tube
6. Counter	1. Iodine	solution (Lugol Reagent)	1 tube
7. Timer	One 7.95	% ethanol (Decolor Reagent)	1 tube
8. Washing bottle	1 bottle	of Safranin stain; 1 bottle	1 tube
9. Oil immersion	of equipment:		1 tube
paint 10. Extra fine pen	1 optical	microscope;	1 unit
11. Alcohol wipes	1 shared	heating plate.	1 unit
12. Waste liquid			
cup 13. Trash can	1		

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including the cover and the question paper) has a total of 4 pages and must be returned in its entirety when submitting the exam.
3. The total time allotted for answering questions in this exam is 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back, but please indicate the question number.

I. Background Introduction

Yogurt is a fermented dairy product that humans have consumed for thousands of years. Numerous scientific studies have shown that the lactic acid bacteria in yogurt are beneficial probiotics for human health. Traditional yogurt contains many different types of lactic acid bacteria, making it a mixed-strain fermented microbial food. Some lactic acid bacteria grow rapidly and produce acidic substances, while others produce various aromatic metabolites, contributing to yogurt's unique flavor. Although there are many brands of yogurt with varying flavors, they typically contain two types of lactic acid bacteria: the *cocciform* *Streptococcus thermophilus* (St) and the rod-shaped *Lactobacillus bulgaricus* (Lb). St and Lb play different roles in the fermentation process, competing yet complementing each other. St grows rapidly and metabolizes to produce organic acids, lowering the pH of the dairy product. The carbon dioxide produced also promotes the growth of Lb. Lb, in turn, breaks down and metabolizes sugars, proteins, and lipids, producing many small molecule metabolites with distinctive aromatic properties. This is the main reason why yogurt is known for its sour taste and unique aroma.

During yogurt fermentation, the populations of St and Lb bacteria fluctuate over time, resulting in changes in their cell ratio. This experiment compares the changes in the ratio of cocci to bacilli in fermentation samples cultured for different durations using staining and microscopic observation. II. Experimental Materials

Yogurt probiotic samples cultured at 42°C for different times (0hr, 3hr, 6hr, 9hr), microscope, oil immersion oil, glass slides, crystal violet stain, iodine solution, 95% ethanol, safranin stain, hot plate, P20 micropipette, lens paper, lint-free paper, counter, timer, wash bottle. III. Experimental Procedure

- Using a p20 micropipette, take 10µl of yogurt and probiotic samples (0hr, 3hr, 6hr, 9hr) and prepare 4 smears at equal intervals on the same glass slide.



- Perform Gram staining as follows: a. After smearing, write

the examinee's number on the slide and take it to the heated plate area, where the teaching assistant will heat it with the heated plate to dry.

Dry it, and then have the teaching assistant return it.

- Stain with crystal violet solution for 1 minute. Then, use a wash bottle to drip water from the top edge of the slide on the front side of the waste liquid cup. Let the water flow slowly over the slide to remove excess stain (referred to as front washing).

- Stain with iodine solution (Lugol Reagent) for 1 minute, then rinse with water from the front of the slide.
- Decolorize with 95% ethanol (Decolor Reagent) for 20 seconds, then rinse with water from the front of the slide.
- Stain with safranin solution (Safrole) for 1 minute, then rinse with water from the front of the slide.
- Place the slide on clean paper and gently blot off any excess moisture with another piece of clean paper.
- Observe the sample. Precautions for using a microscope:

First, use a scanning microscope (4X) to locate the area to be observed. Then, gradually adjust to 40x to confirm the clarity. Turn the objective lens to the left, add one small drop of oil immersion fluid, and then rotate to 100x for observation. 3. Select any field of view and count the number of cocci and bacilli cells. Count three fields of view for each smear. Take bacterial cells...

- Calculate the ratio of cocci

to bacilli. 5. Draw a ratio curve on graph paper. 6.

After completing this test, confirm that the slide has

been written with your test number and placed on clean paper. The teaching assistant will collect it after the test. 7. After the experiment is completed: (1)

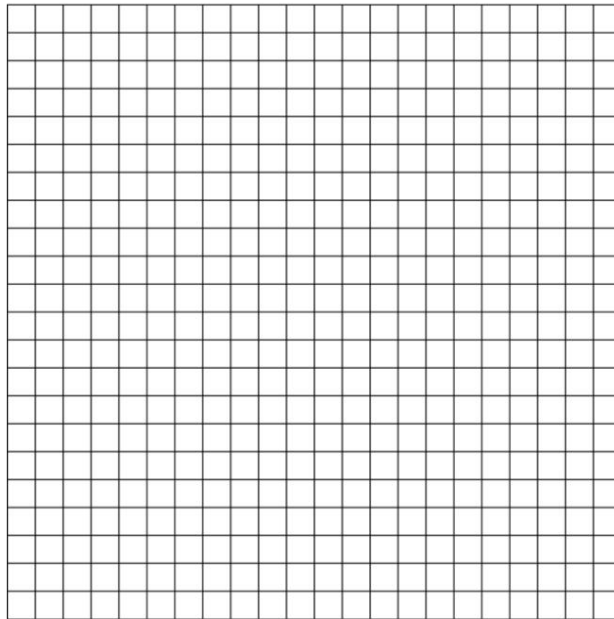
Zero the counter. (2) Open the alcohol wipe, squeeze out the excess alcohol, wipe the excess oil stains on the 40X and 100X objectives, and then wipe them dry with lens paper. (3) Tidy up the table (Please make sure you finish! Otherwise, points will be deducted).

IV. Experimental Results and

Discussion 1. Data Completion

(30%) Hours		0hr			3hr			6hr			9hr		
Cell number													
Cocci	Count 3 fields of view												
Cell number	On average, 3 fields of view												
Number of bacillus cells	are counted per field of view.												
	Average number of												
cocci:bacteria ratio per field of view													

2. Cartography (20%)



3. Please describe your experimental results. (25%)

V. Problems

1. Besides incubation time, what other factors do you think affect the growth and decline of cocci and bacilli? Please list them and discuss them. (15%)

2. Are St and Lb Gram-positive or Gram-negative bacteria, respectively? (5%)

3. What are probiotics? Besides lactic acid bacteria, please give an example of a probiotic. (5%)